INCH-POUND

MIL-DTL-17587/6(SH)
28 June 2007

DETAIL SPECIFICATION SHEET

CIRCUIT BREAKERS, ACB, LOW VOLTAGE, ELECTRIC POWER, AIR, REMOVABLE CONSTRUCTION ACB-4000HR AND ACB-4010HR

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-17587.

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REQUIREMENTS:

TABLE I. Ratings/features of circuit breakers.

Circuit breaker type	**ACB-4000HR (PIN 1)**	**ACB-4000HR (PIN 2)**	**ACB-4010HR**
Rated continuous current (amperes)	4000	4000	4000
Rated maximum voltage (AC volts) / and frequency (60 hertz)	500	500	500
Frequency (hertz)	60	60	60
Maximum interrupting rating current (symmetrical amperes)	85,000	85,000	85,000
Rated short-time current /2 (amperes)	85,000	85,000	85,000
Rated short time current duration (seconds)	0.75	0.75	0.75
Operating mechanism – manual	STD	STD	STD
Operating mechanism – electric	STD	STD	STD
Maximum weight (lbs) – circuit breaker (removable component)	726	658	360
Maximum weight (lbs) – cradle (stationary component)	481	495	410
Number of poles	3	3	3
Number of overcurrent coils or solid-state sensors	2	3	3
Attachments			
Electric closing mechanism	STD	STD	STD
Auxiliary switches (number of contacts)	8	10	12
Undervoltage lockout device	STD	STD	STD
Mechanical position indicator	STD	STD	STD
Secondary disconnecting devices	STD	STD	STD
Cell switch	STD	STD	STD
Indicator light	STD	OPT	STD
Hold-closed mechanism	STD	OPT	STD
Undervoltage release device	STD	OPT	STD
Shunt trip device	STD	STD	STD
Operations counter	STD	STD	STD
Trip on power up (TOPU) device	N/A	N/A	OPT
Rack-in/rack-out position indicator	STD	STD	STD

NOTES:

/1 Voltage ratings apply to main current-carrying parts. Operating attachments shall be of voltages specified in 6.2 in MIL-DTL-17587.

/2 Ampere values apply to circuit breakers with overcurrent coils or sensors of maximum current rating within the circuit breaker type.

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TABLE II. Available continuous current ratings and pickup current settings for overcurrent coils and sensors.

Type	Coil or sensor rating /1 (amperes)	Long-time pickup setting (amperes)			Short-time pickup setting (amperes)					Maximum instantaneous trip /2 (amperes)
ACB-4000HR (PIN 1)	2000	3200	4000	4800	4000	4800	6000	8000	----	68000
	2400	3600	4800	6000	4800	6000	8000	----	----	68000
	2800	4200	5600	6600	5600	7000	8400	----	----	68000
	3200	4800	6400	8000	6400	8000	9600	----	----	68000
	4000	6000	8000	10000	8000	10000	12000	----	----	68000
ACB-4000HR (PIN 2)	2000	3000	4000	5000	4000	5000	6000	8000	10000	80000
	2400	3600	4800	6000	4800	6000	7200	9600	12000	80000
	2800	4200	5600	7000	5600	7000	8400	11200	14000	80000
	3200	4800	6400	8000	6400	8000	9600	12800	16000	80000
	4000	6000	8000	10000	8000	10000	12000	16000	20000	80000
ACB-4010HR	2000	3000	4000	5000	4000	5000	6000	8000	10000	80000
	2400	3600	4800	6000	4800	6000	7200	9600	12000	80000
	2800	4200	5600	7000	5600	7000	8400	11200	14000	80000
	3200	4800	6400	8000	6400	8000	9600	12800	16000	80000
	4000	6000	8000	10000	8000	10000	12000	16000	20000	80000

NOTES:

/1 Continuous current rating.

/2 The instantaneous trip device shall be set at the place of manufacture from five times the coil or sensor rating to the maximum pickup setting specified for mechanical coils and 20 times sensor rating for solid-state sensors with a default at maximum listed rating.

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TABLE III. Electrical endurance and temperature rise test requirements.

Electrical endurance test:

Type	Total number of close-open cycles	Amperes on circuit breaker close	Amperes on circuit breaker open	Maximum AC power factor (lagging) X/R
ACB-4000HR (PIN 1)	1,000	4,000	4,000	0.8
ACB-4000HR (PIN 2)	1,000			0.8
ACB-4010HR	1,000			0.8

Temperature rise test:

Type	Continuous current rating	Cable quantity per terminal /1	Cable size	Bus bar quantity per phase	Bus bar size (inches) /2
ACB-4000HR (PIN 1)	4,000	10 /3	400 MCM	4	6.0 x ¼
ACB-4000HR (PIN 2)					
ACB-4010HR					

NOTES:

/1 Information regarding cable is contained in MIL-DTL-915. If 3-phase cable is used for this test, the cable jacket shall be stripped for a minimum of 2 feet.

/2 Information regarding bus bars is contained in MIL-DTL-16036.

/3 Connect with bus adapter of minimum length.

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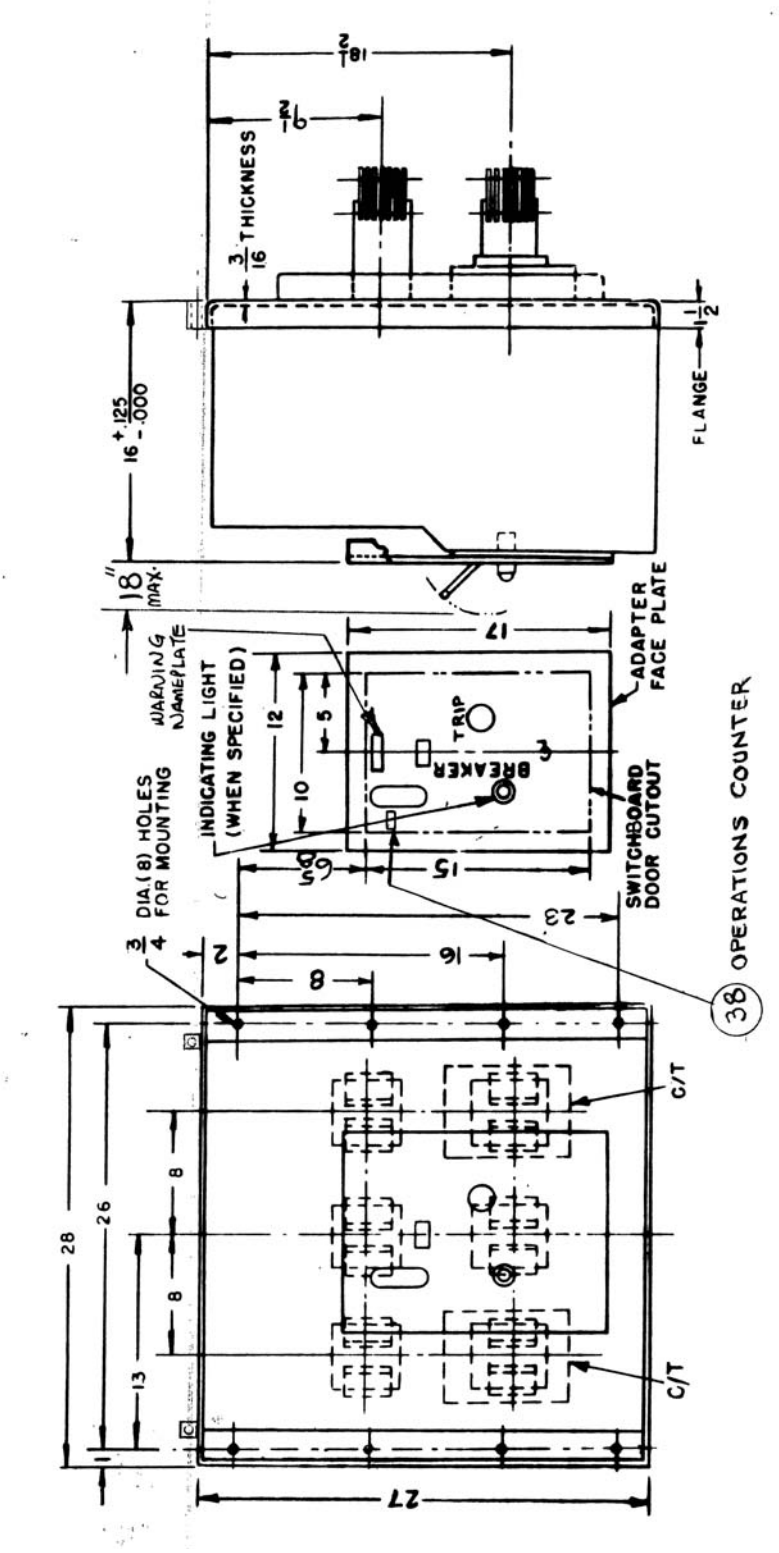


FIGURE 1. Outline and mounting dimensions for circuit breaker type ACB-4000HR (PIN 1).

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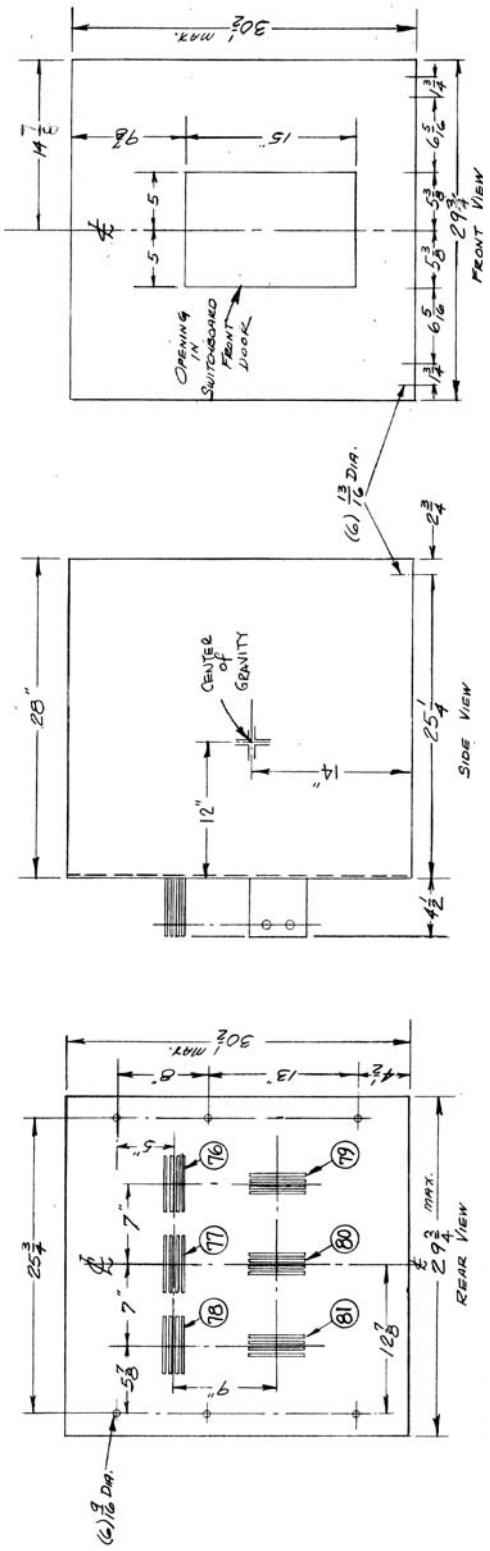


FIGURE 2. Cradle outline and mounting dimensions for circuit breaker type ACB-4000HR (PIN 1).

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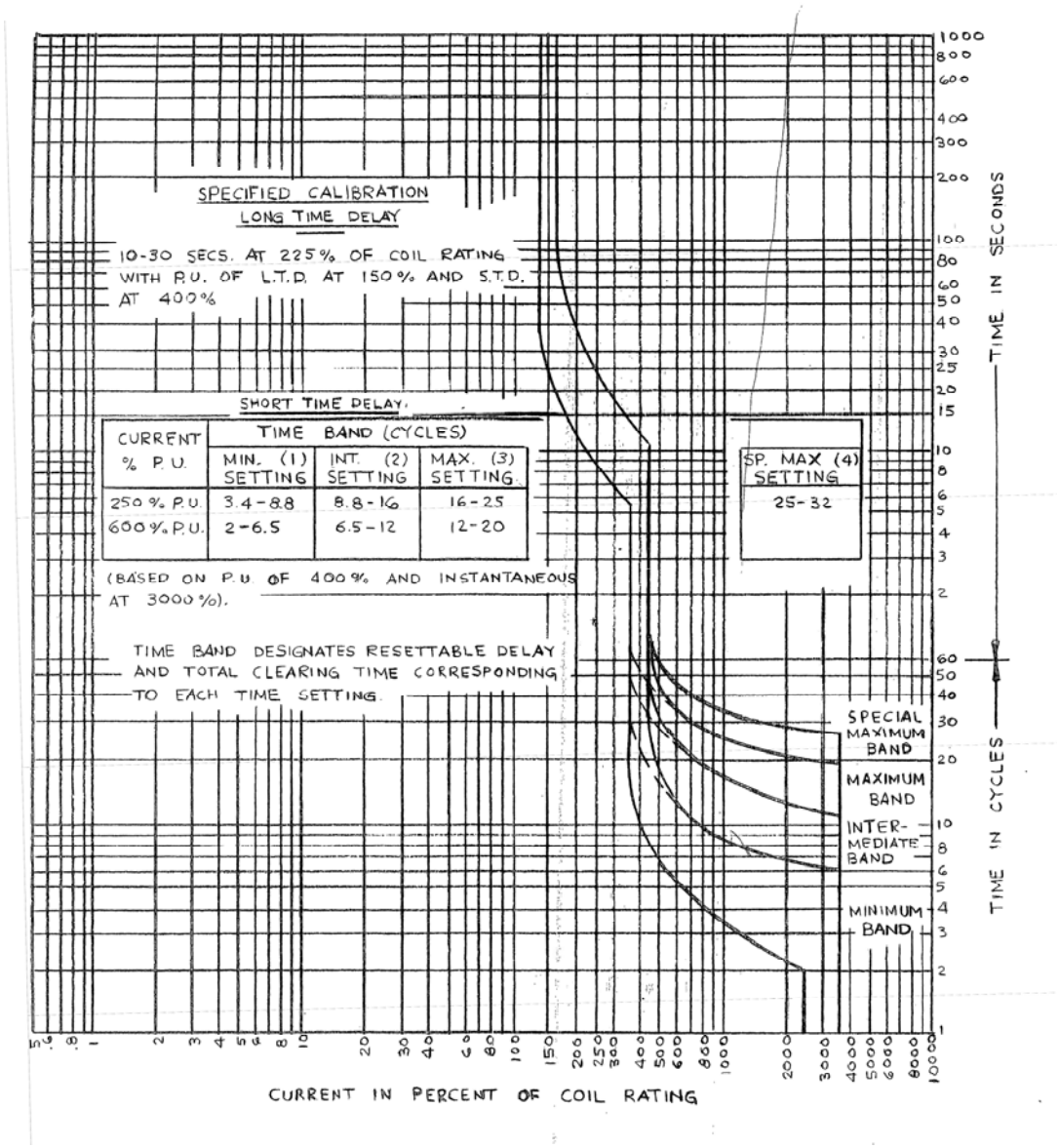


FIGURE 3. Time-current characteristic curve for circuit breaker type ACB-4000HR (PIN 1).

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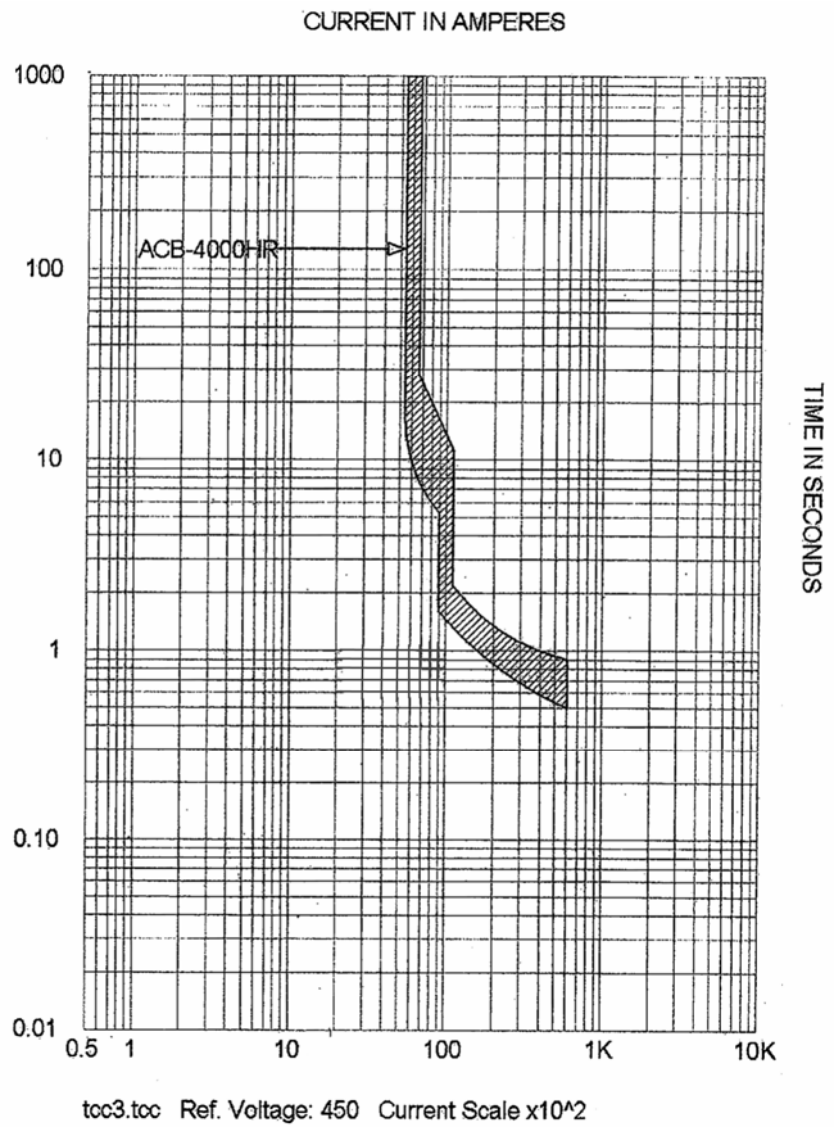


FIGURE 4. Short-time delay band Special 4 characteristics curve for circuit breaker type ACB-4000HR (PIN 1).

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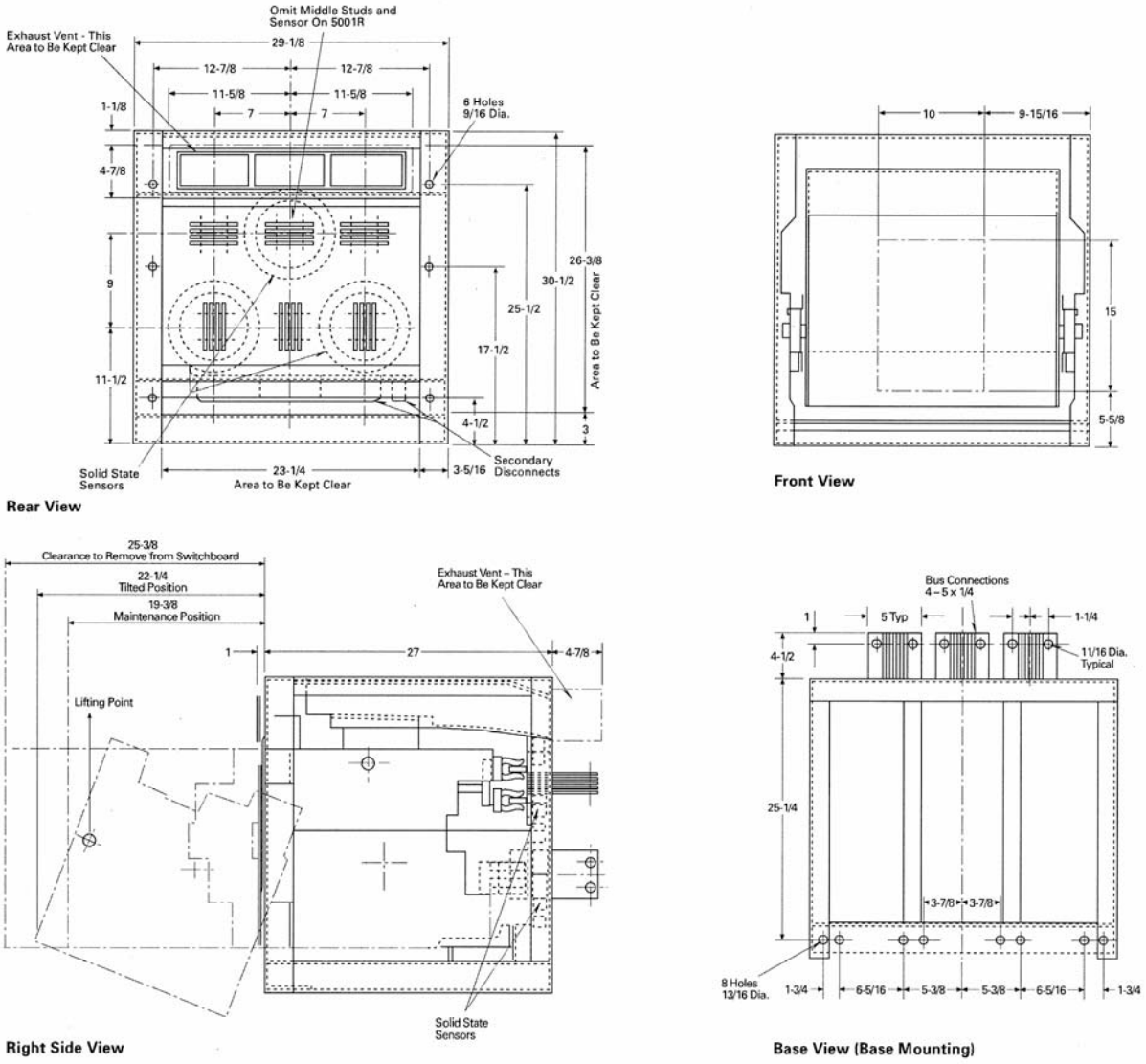


FIGURE 5. Outline and mounting dimensions for circuit breaker type ACB-4000HR (PIN 2).

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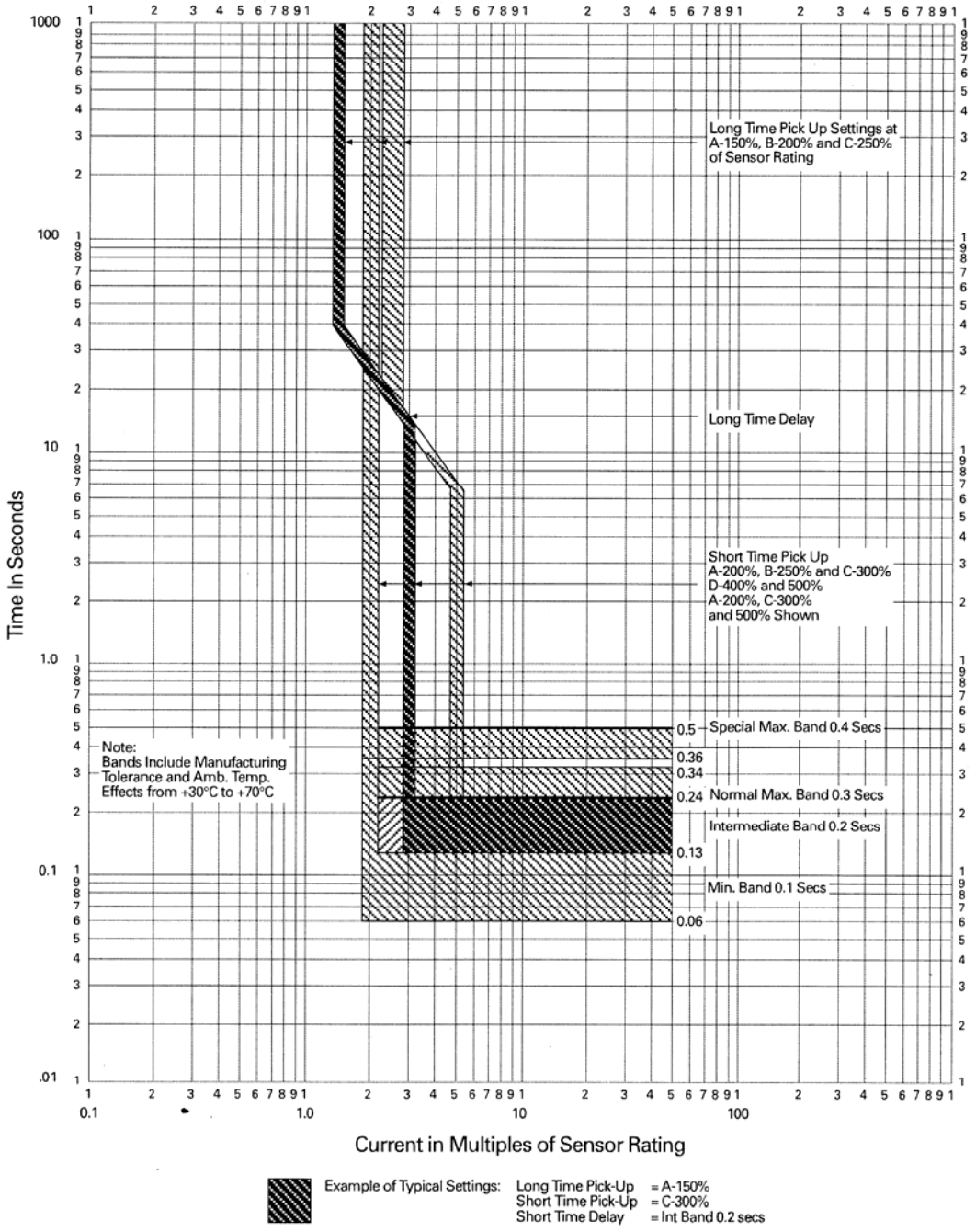


FIGURE 6. Time-current characteristic curve for circuit breaker type ACB-4000HR (PIN 2).

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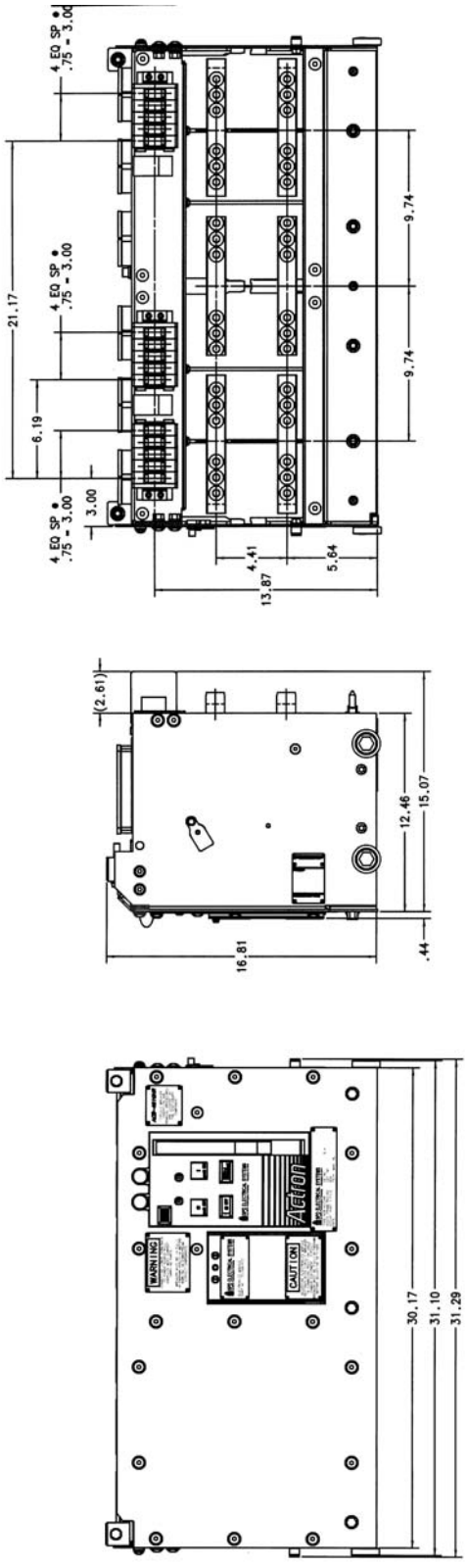


FIGURE 7. Outline and mounting dimensions for circuit breaker type ACB-4010HR.

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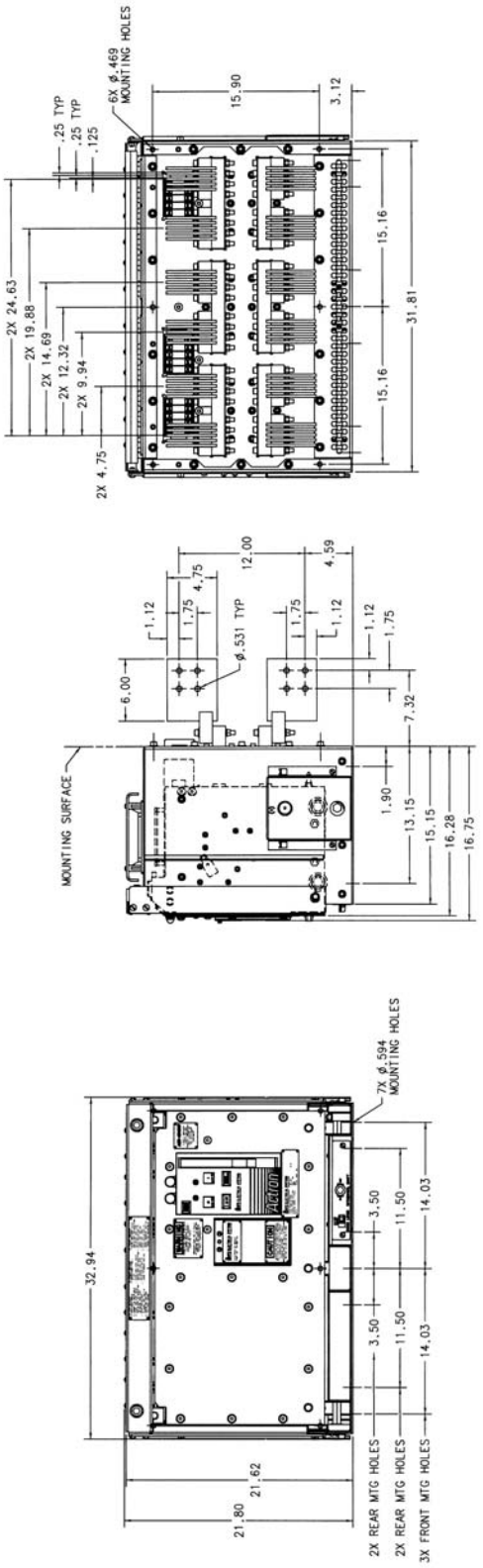


FIGURE 8. Outline and mounting dimensions for circuit breaker type ACB-4010HR installed in cradle.

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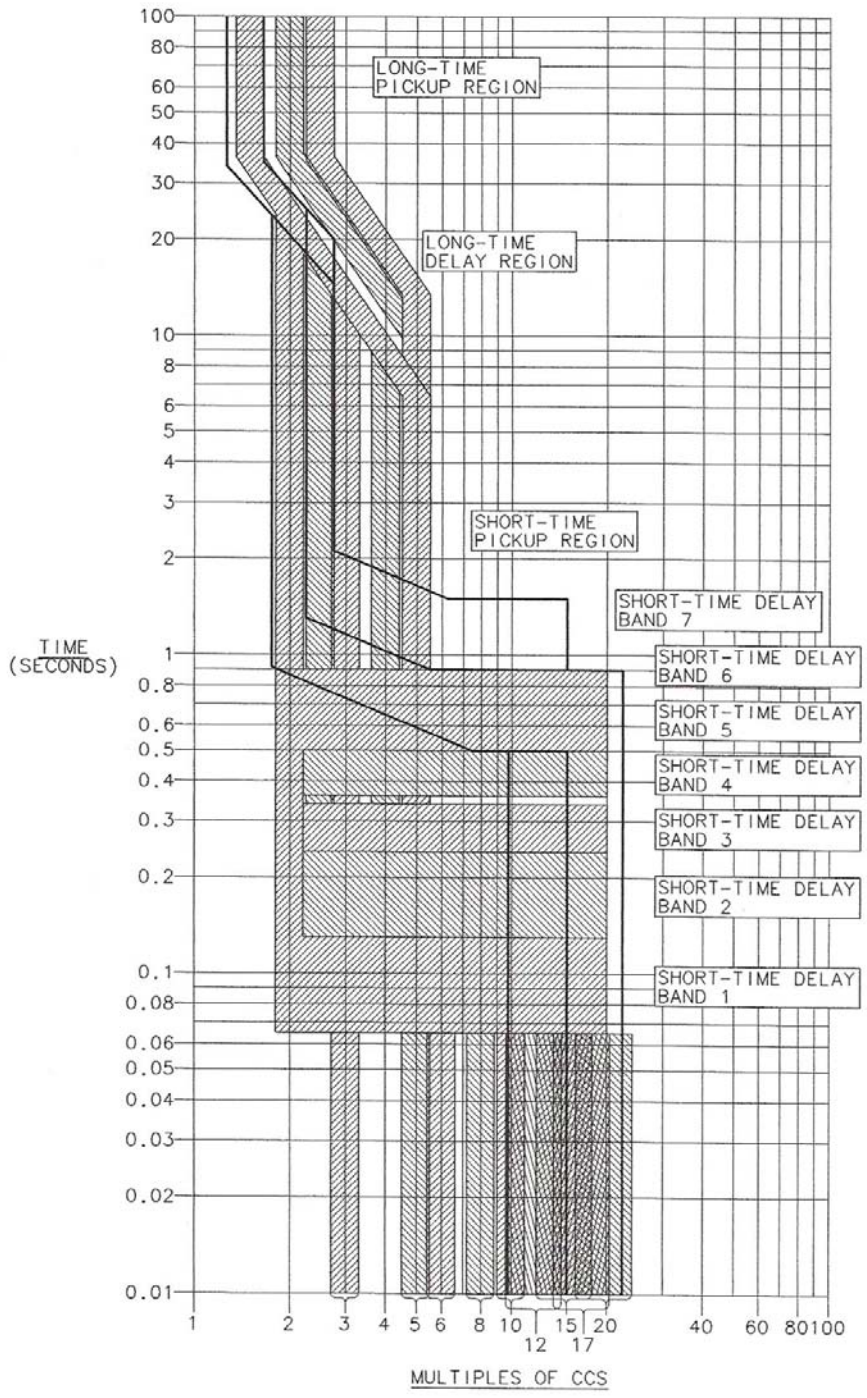


FIGURE 9. Time-current characteristic curve for circuit breaker type ACB-4010HR.

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ADJUSTABLE RATINGS AND CONTINUOUS CURRENT SETTINGS (CCS)				
2000A	2400A	2800A	3200A	4000A

ADJUSTABLE LONG-TIME DELAY PICKUP SETTINGS (IN MULTIPLES OF CONTINUOUS CURRENT SETTINGS)		
1.5X	2.0X	2.5X

ADJUSTABLE SHORT-TIME DELAY PICKUP SETTINGS (IN MULTIPLES OF CONTINUOUS CURRENT SETTINGS)				
2.0X	2.5X	3.0X	4.0X	5.0X

ADJUSTABLE INSTANTANEOUS PICKUP SETTINGS (IN MULTIPLES OF 4000 AMPERE FRAME RATING)								
3X	5X	6X	8X	10X	12X	15X	17X	20X
12000	20000	24000	32000	40000	48000	60000	68000	80000

ADJUSTABLE SHORT-TIME DELAY SETTINGS	
SHORT-TIME DELAY BAND	DELAY (SECONDS)
1	.065-.130
2	.130-.240
3	.240-.340
4	.360-.500
5	.500-.900
6	.500-1.50
7	.900-2.05

PER MIL-DTL-17587

LONG-TIME AND SHORT-TIME PICKUP CURRENTS ARE ACCURATE TO ±10%. INSTANTANEOUS PICKUP CURRENTS ARE ACCURATE TO:

±10% FOR PICKUPS ≤ 10X FRAME RATING
±20% FOR PICKUPS > 10X FRAME RATING

LONG-TIME AND SHORT-TIME CALIBRATION ARE GUARANTEED OVER TEMPERATURE RANGE DEFINED BY SWITCHGEAR AMBIENT OF 10°C TO 50°C.

INSTANTANEOUS TRIP POINTS ARE GUARANTEED AT 25°C ONLY. CONSULT FACTORY FOR ACCURACY AT HIGHER TEMPERATURE.

TRIP TIMES ARE FROM INITIATION OF SYMMETRICAL SINUSOIDAL CURRENT TO EXTINGUISHING OF ARC AT RATED VOLTAGE.

FIGURE 9. Time-current characteristic curve for circuit breaker type ACB-4010HR - Continued.

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Preparing Activity
Navy – SH
(Project 5925-N184-006)

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